

Notes: Rostek & Yoder (RES, 2026)

Reallocative Auctions and Core Selection

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1 Big picture

- **Question.** When can an auction that reallocates goods among bidders select a core outcome? The paper studies auctions in which participants can both buy and sell, rather than only buy from the auctioneer.
- **Main departure from standard auction theory.** In one-sided auctions, gross substitutability of goods is the central condition behind core selection. In reallocative auctions, substitutability among goods is neither sufficient nor necessary.
- **Main thesis.** The key obstacle is *heterogeneity among bidders*: heterogeneity in pre-auction marginal utilities and heterogeneity in substitution patterns.
- **Why reallocation changes the problem.** When bidders can sell as well as buy, the efficient allocation can require the auctioneer to facilitate trade among bidders. The Vickrey rule may subsidize those trades, producing revenue that is not competitive.
- **Core selection intuition.** Core selection is desirable because it ensures efficient allocation, competitive revenue, resistance to shill bidding, and monotonicity of revenue in participation. In reallocative environments the relevant notion is often *bidder-core selection*, which abstracts from whether revenue is positive and focuses on revenue competitiveness.
- **First main result.** If there exists a basis of packages that bidders view as substitutes, and the efficient allocation gives every bidder nonnegative quantities of those packages, then the Vickrey auction is bidder-core selecting.
- **Second main result.** Such packages exist when bidders' pre-auction marginal utilities are equal and their substitution patterns commute, or equivalently are simultaneously diagonalizable. The common eigenvectors define the right packages.
- **Negative results.** If either pre-auction marginal utilities or substitution patterns are sufficiently heterogeneous, bidder-core selection can fail, even with substitutable goods.
- **Revenue comparison.** A uniform-price auction is not efficient but avoids subsidizing reallocation. Vickrey revenue is higher when heterogeneity is small; the uniform-price auction can dominate when heterogeneity is large.
- **Implementation lesson.** The right package design can reduce complexity: independent auctions for eigenvector packages can replicate the full reallocative Vickrey or uniform-price auction under homogeneity conditions.
- **Economic takeaway.** The paper shifts the design question from “Are the goods substitutes?” to “Are the bidders sufficiently similar in their marginal values and substitution patterns, and does the auctioneer's quantity create the gains from trade?”

2 Data, environment, and model primitives

This is a theoretical paper. No new empirical data are generated. The “data” of the model are primitives of the reallocation auction environment: bidders, quantities, valuations, feasible allocations, and coalitional values.

2.1 Auctioneer, bidders, and goods

- There is an auctioneer, denoted a , and a set I of potential bidders.
- A finite subset $X \subseteq I$ participates in a given auction, with $|X| \geq 2$.
- There are K perfectly divisible goods.
- The auctioneer has a quantity vector

$$\bar{q} \in \mathbb{R}^K.$$

- The auctioneer values the goods at zero; his payoff is the sum of transfers received from bidders.
- Bidder i receives net quantity $q_i \in \mathbb{R}^K$ and pays transfer $x_i \in \mathbb{R}$.
- A positive component of q_i means the bidder buys that good; a negative component means the bidder sells that good.

Interpretation. Unlike one-sided auctions, the feasible set does not impose $q_i \geq 0$. Bidders can be net buyers, net sellers, or buyers of some goods and sellers of others. This is exactly what creates the new core-selection problem.

2.2 Valuations and feasibility

Each bidder has a valuation

$$u_i : \mathbb{R}^K \rightarrow \mathbb{R}, \quad u_i(0) = 0.$$

The payoff of bidder i is

$$u_i(q_i) - x_i.$$

The paper assumes u_i is twice continuously differentiable and strictly concave. The only feasibility condition is

$$\sum_{i \in X} q_i = \bar{q}.$$

Interpretation. Feasibility says that the auctioneer’s quantity and the bidders’ trades must clear. Since negative quantities are allowed, the allocation can reallocate goods among bidders even when the auctioneer supplies little or nothing.

2.3 Surplus function and efficient allocation

For a coalition of bidders $Z \subseteq I$, define the surplus function

$$v(Z, q) = \max_{\{q_i\}_{i \in Z}} \sum_{i \in Z} u_i(q_i) \quad \text{s.t.} \quad \sum_{i \in Z} q_i = q. \quad (1)$$

The efficient allocation for coalition Z and quantity q is

$$\{q_i^e(Z, q)\}_{i \in Z} = \arg \max_{\{q_i\}_{i \in Z}} \sum_{i \in Z} u_i(q_i) \quad \text{s.t.} \quad \sum_{i \in Z} q_i = q. \quad (2)$$

Interpretation. $v(Z, q)$ is the maximum surplus a coalition can create if it controls quantity q . The core conditions are written in terms of whether coalitions can create more surplus on their own than they receive in the auction payoff profile.

2.4 Coalitional value function

For any coalition $Z \subseteq I \cup \{a\}$, define

$$V(Z, \bar{q}) = \begin{cases} v(Z \setminus \{a\}, \bar{q}), & a \in Z, \\ v(Z, 0), & a \notin Z. \end{cases} \quad (3)$$

Interpretation. If the coalition contains the auctioneer, it controls the auctioneer's quantity $\bar{q}$. If it does not contain the auctioneer, bidders can only trade among themselves, so the net quantity is zero. This second case is the critical reallocation feature missing from one-sided auctions.

2.5 Core and bidder-core

A payoff vector $\pi \in \mathbb{R}^{X \cup \{a\}}$ is in the core if it is feasible and no coalition can block it:

$$\sum_{i \in X \cup \{a\}} \pi_i = V(X \cup \{a\}, \bar{q}), \quad (4)$$

$$\sum_{i \in Z} \pi_i \geq V(Z, \bar{q}) \quad \text{for every } Z \subseteq X \cup \{a\}. \quad (5)$$

If a payoff profile is unblocked by all coalitions except the auctioneer alone, $Z = \{a\}$, it is in the *bidder-core*.

Interpretation. Bidder-core selection focuses on competitive revenue rather than on whether revenue is strictly nonnegative. This matters because reallocation Vickrey auctions can naturally have negative revenue, as in procurement settings, without necessarily having uncompetitive revenue.

2.6 Quadratic valuations and substitution patterns

The paper uses quadratic examples and revenue comparisons. A quadratic valuation is

$$u_i(q_i) = \theta'_i q_i - \frac{1}{2} q'_i S_i q_i, \quad (6)$$

where S_i is positive definite. The Hessian is

$$D^2 u_i(q_i) = -S_i.$$

Interpretation. The matrix S_i encodes bidder i 's substitution pattern. If different bidders' S_i matrices have different eigenvectors, they disagree about the natural package directions. This disagreement is the source of heterogeneity that can make core selection fail.

2.7 No empirical tables

The article is a theory paper. It contains no empirical tables. Its main visual evidence consists of two figures illustrating Theorem 1's allocation condition and the three motivating examples.

3 Models and theoretical specifications

3.1 The Vickrey auction and dominant strategy equilibrium

The Vickrey implementation solicits inverse demand schedules. Bidder i submits a bid function $b_i : R^K \rightarrow M$. The market-clearing price $p^*(b)$ solves

$$\bar{q} = \sum_{i \in X} b_i^{-1}(p^*(b)).$$

Bidder i receives

$$q_i^*(b) = b_i^{-1}(p^*(b)).$$

Proposition 1 states that truthful bidding is weakly dominant:

$$b_i^V(q_i) = \nabla u_i(q_i). \quad (7)$$

The Vickrey auction implements the efficient allocation:

$$\{q_i^*(b^V)\}_{i \in X} = \{q_i^e(X, \bar{q})\}_{i \in X}. \quad (8)$$

Bidder i 's Vickrey payoff is his marginal contribution:

$$\pi_i^V(X) = v(X, \bar{q}) - v(X \setminus \{i\}, \bar{q}). \quad (9)$$

The auctioneer receives the residual surplus

$$\pi_a^V(X) = v(X, \bar{q}) - \sum_{i \in X} \pi_i^V(X). \quad (10)$$

Interpretation. The Vickrey auction is the natural benchmark because any incentive-compatible, core-selecting mechanism on a convex domain must be outcome-equivalent to Vickrey. Thus the question becomes: When is the Vickrey payoff profile in the bidder-core?

3.2 Why bidder-core selection is the relevant competitiveness condition

With reallocation, bidder-only coalitions can create value by trading among themselves. Proposition 2 shows these coalitions never block the Vickrey payoff profile: participation in the reallocative Vickrey auction is coalitionally rational for bidders. Therefore blocking coalitions relevant for revenue competitiveness must include the auctioneer.

Corollary 1 then states:

- Vickrey revenue is competitive if and only if the Vickrey outcome is bidder-core selecting.
- Vickrey revenue is competitive and nonnegative if and only if the Vickrey outcome is core selecting.

Interpretation. The paper separates two ideas: (i) revenue being nonnegative, and (ii) revenue being competitive against deviations by coalitions. Bidder-core selection captures the second idea.

3.3 Goods substitutability

Bidder i 's demand at price p is

$$d_i(p) = \arg \max_{q_i} \{u_i(q_i) - p \cdot q_i\}. \quad (11)$$

Goods are substitutes if, when the price of one good rises, demand for every other good weakly rises.

Interpretation. In one-sided environments this condition is enough for core selection. The paper shows that this classical condition fails to characterize core selection once bidders can buy and sell.

3.4 Example 1: no core selection with one good and positive revenue

The auctioneer sells $\bar{q} = 1$ unit of a single good. Three bidders have valuations

$$u_1(q_1) = 4q_1 - \frac{1}{2}q_1^2, \quad u_2(q_2) = 4q_2 - \frac{1}{2}q_2^2, \quad u_3(q_3) = 2q_3 - \frac{1}{2}q_3^2. \quad (12)$$

The efficient allocation gives one unit to bidders 1 and 2, and makes bidder 3 sell one unit:

$$q_1^e = q_2^e = 1, \quad q_3^e = -1.$$

Interpretation. Even with one good, and hence no complementarity issue among goods, the Vickrey auction may subsidize reallocation from a low-marginal-value bidder to a high-marginal-value bidder. A coalition containing the auctioneer and bidder 1 can block, so core selection fails.

3.5 Example 2: core selection with both complements and substitutes

The auctioneer sells $\bar{q} = [1, 1]'$. Bidder 1 views the two goods as complements, while bidders 2 and 3 view them as substitutes. Nevertheless, all bidders' Hessian matrices have the same eigenvectors:

$$\phi_1 = [1, 1]', \quad \phi_2 = [-1, 1]'$$

The Vickrey auction is bidder-core selecting.

Interpretation. Complementarity among the original goods is not what matters. The relevant question is whether there are packages that all bidders treat as substitutes. In this example, the common eigenvectors are the right packages.

3.6 Example 3: no core selection despite substitutable goods

In Example 3, bidders have the same marginal utility at zero and the auctioneer sells nonnegative quantities of goods. Goods are substitutes for all bidders. Yet the Vickrey payoff profile is blocked because bidders' substitution patterns are heterogeneous.

Interpretation. Substitutability of the named goods is not sufficient. The problem is that new supply of one good can create efficient trades in another good among bidders. The Vickrey auction subsidizes this induced reallocation.

3.7 Package substitutability

Let $\Phi = \{\phi_1, \dots, \phi_K\} \subset \mathbb{R}^K$ be a set of K linearly independent packages, and let T_Φ be the matrix whose columns are the packages. Package substitutability means that in the transformed basis, when the price of package ϕ_k increases, demand for the other packages weakly increases.

Interpretation. A package is a direction in the space of goods. It may be a buying package, a selling package, or a switching package that buys some goods while selling others. The basis change asks whether bidders see these directions as substitutes.

3.8 Theorem 1: Core Selection - Package Substitutability and Allocations

Theorem 1 gives sufficient conditions for bidder-core selection. Suppose:

1. Packages Φ are substitutes.
2. For every relevant participant set $Z \subseteq I$ and each bidder $i \in Z$, the efficient allocation gives bidder i nonnegative quantities of each package:

$$T_{\Phi}^{-1} q_i^e(Z, \bar{q}) \geq 0. \quad (13)$$

Then the Vickrey auction is bidder-core selecting for every set of participating bidders.

Interpretation. Package substitutability alone is not enough. The allocation condition is what turns a reallocation auction for goods into a one-sided auction for packages: each bidder is effectively receiving nonnegative amounts of the relevant packages rather than selling some package directions to others.

3.9 Theorem 2: Core Selection - Package Substitutability and Heterogeneity

Theorem 2 gives primitive conditions ensuring Theorem 1. Suppose:

1. Bidders' pre-auction allocations are efficient:

$$\nabla u_i(0) = \nabla u_j(0) \quad \text{for all } i, j \in I. \quad (14)$$

2. Bidders' substitution patterns commute:

$$D^2 u_i(q_i) D^2 u_j(q_j) = D^2 u_j(q_j) D^2 u_i(q_i). \quad (15)$$

Then the substitution patterns share a common orthonormal eigenbasis Φ such that packages Φ are substitutes and efficient allocations give each bidder positive quantities of each package. Hence the Vickrey auction is core selecting for every participant set.

Interpretation. Commutativity is a formal homogeneity condition. It says bidders may value goods differently, but they agree on the independent package directions. Equal pre-auction marginal utility says there are no pre-existing gains from trade before the auctioneer's quantity arrives.

3.10 Maximal-domain negative results

The paper proves that the two heterogeneity restrictions are not cosmetic.

- **Proposition 3.** For any valuation profile, there are pre-auction allocations that make the Vickrey auction fail bidder-core selection for some participant set.
- **Proposition 4.** If pre-auction marginal utilities differ, then some auctioneer quantity $\bar{q}$ makes the Vickrey auction fail bidder-core selection.
- **Proposition 5.** Sufficient heterogeneity in substitution patterns can make bidder-core selection fail even when the pre-auction allocation is efficient.

Interpretation. The failure is not a boundary case. If bidder heterogeneity is allowed to be arbitrary, a designer cannot rely on Vickrey/core selection in a robust domain.

3.11 Uniform-price auction and Theorem 3

The uniform-price auction uses the same allocation rule as Vickrey but charges

$$t_i^U(b) = p^*(b) \cdot q_i^*(b). \quad (16)$$

For quadratic valuations with a common substitution pattern $-S$, Proposition 6 characterizes a unique linear ex post equilibrium:

$$b_i^U(q_i) = \theta_i - \frac{|X| - 1}{|X| - 2} S q_i. \quad (17)$$

Theorem 3 compares expected revenue under i.i.d. quadratic valuations. Let $\Phi = \{\phi_k\}_{k=1}^K$ be the eigenbasis of S with eigenvalues $\{\lambda_k\}_{k=1}^K$. Then expected revenue is higher in the Vickrey auction than in the uniform-price auction if and only if

$$\sum_{k=1}^K \frac{1}{\lambda_k} \text{Var}_F(\phi_k \cdot \theta) < \frac{1}{(N-1)(N-2)} \bar{q}' S \bar{q}. \quad (18)$$

Interpretation. The left side is expected heterogeneity in bidders' marginal utilities along package directions. The right side is the surplus/revenue potential created by the auctioneer's quantity. Vickrey beats UPA when heterogeneity is small relative to the quantity the auctioneer contributes.

3.12 Implementation: eigenvector packages and split auctions

Proposition 7 shows that if bidders' substitution patterns commute, then independent Vickrey auctions for eigenvector packages produce the same allocation and payoffs as the full joint Vickrey auction. With identical quadratic substitution patterns and at least three bidders, the same equivalence holds for independent uniform-price package auctions.

Proposition 8 shows that splitting a Vickrey auction into forward and reverse auctions is generally not innocuous: truthful bidding is no longer a weakly dominant strategy in a split Vickrey auction. By contrast, a split uniform-price auction is outcome-equivalent to a single reallocative UPA because payments depend only on the market-clearing price and allocated quantities.

Interpretation. Eigenvector packaging is a complexity-reduction tool that respects the economic structure of preferences. Split auctions are less benign for Vickrey because they change residual supply curves and therefore change payments.

4 Identification and theoretical strategy

4.1 What plays the role of identification

This is not an empirical paper. The equivalent of an identification strategy is a sequence of theoretical comparisons that isolate which features of a reallocative environment determine core selection.

- The model first defines the benchmark Vickrey outcome, because any incentive-compatible core-selecting mechanism must be Vickrey-equivalent on the relevant domains.
- It then asks whether the Vickrey payoff profile lies in the core or bidder-core.
- Examples 1-3 isolate three candidate explanations: one-good reallocation, complementarity, and substitutability.
- Theorems 1-2 then identify the actual sufficient structure: package substitutability plus nonnegative package allocations, generated by low heterogeneity.
- Propositions 3-5 show the result is tight in a maximal-domain sense: removing those homogeneity restrictions permits failure.

Interpretation. The paper's theoretical design works like a set of counterexamples and converse-style results. The authors first show that old conditions fail, then identify new sufficient conditions, then show that the new restrictions cannot be dropped broadly.

4.2 Step 1: Separate full core from bidder-core

The paper first distinguishes two issues:

- whether revenue is competitive against coalitions that include bidders and the auctioneer;
- whether revenue is also nonnegative.

Bidder-core selection focuses on competitiveness and abstracts from a block by the auctioneer alone.

Interpretation. This distinction is essential because reallocative auctions can naturally have negative revenue even when payments are competitive. Without this separation, the analysis would mix procurement-style budget balance with revenue competitiveness.

4.3 Step 2: Use examples to reject the one-sided intuition

The examples identify failures of old intuitions:

- Example 1: substitutability is trivial with one good, but core selection fails.
- Example 2: goods can be complements for one bidder, but core selection succeeds.
- Example 3: goods can be substitutes for everyone and pre-auction marginal utilities equal, but core selection fails because substitution patterns differ.

Interpretation. The examples demonstrate that the problem is not complementarity among the original goods. The problem is whether efficient reallocation requires subsidized bidder-to-bidder trades.

4.4 Step 3: Change the basis from goods to packages

Theorem 1 transforms the space of goods into a space of packages. The relevant objects are not necessarily original goods such as spectrum licenses in cities A and B, but directions such as:

- buying more of both cities' licenses;
- switching spectrum from one city to another;
- selling one package direction while buying another.

Interpretation. A good package basis aligns the auction design with bidders' substitution behavior. This is why package design can matter as much as payment-rule design.

4.5 Step 4: Relate package existence to primitive heterogeneity

Theorem 2 moves from package conditions to primitives. If Hessian matrices commute, they have a common eigenbasis; if pre-auction marginal utilities agree, there are no pre-existing gains from trade that the auction must subsidize.

Interpretation. The result turns an abstract package condition into a checkable economic condition: Are bidders similar in their substitution patterns and marginal values at initial allocations?

4.6 Step 5: Compare Vickrey and uniform price under controlled heterogeneity

Theorem 3 imposes i.i.d. quadratic valuations with a common substitution pattern. This lets the paper express revenue differences in terms of heterogeneity variance and quantity supplied.

Interpretation. The revenue comparison is not a general ranking of mechanisms. It shows that the same heterogeneity that undermines core selection also makes UPA more attractive relative to Vickrey.

4.7 Step 6: Translate theory into implementation

Proposition 7 provides a way to decentralize bidding into independent package auctions when substitution patterns align. Proposition 8 warns that splitting a Vickrey auction into forward and reverse components can change incentives.

Interpretation. The implementation section ties the theory to market design practice, especially spectrum auctions and natural-resource reallocations.

4.8 Main assumptions and threats to interpretation

- Goods are perfectly divisible. Indivisibilities would create additional equilibrium and exposure issues.
- Valuations are quasilinear and strictly concave.
- The paper focuses on Vickrey and UPA benchmarks, not all possible practical payment rules.
- Some results use quadratic valuations, especially the UPA revenue comparison.
- The package approach depends on whether designers can estimate or elicit common substitution directions.
- The model abstracts from dynamic participation, budget constraints, collusion beyond core blocking, and incomplete information about valuations except in the expected-revenue section.

Interpretation. These assumptions make the mechanism-design logic transparent. The broad message is robust conceptually, but practical applications would need estimates of bidder heterogeneity and substitution patterns.

5 Tables and figures

5.1 Figure 1: Theorem 1's allocation condition

Read:

- The figure has two panels, each with two bidders and two goods.
- The vectors ϕ_1 and ϕ_2 are package directions.
- The shaded cone represents combinations of packages that can be allocated in nonnegative quantities.
- In the left panel, both bidders' efficient allocations lie in the positive cone generated by ϕ_1 and ϕ_2 .
- In the right panel, one bidder's allocation falls outside the cone, so Theorem 1's allocation condition fails.

Interpretation. The figure makes the allocation condition visual. It is not enough for packages to be substitutes; the efficient allocation must use those packages as positive quantities for all bidders. If some bidder must effectively sell a package direction to another bidder, the one-sided analogy breaks down.

Economic message. The figure shows why core selection in reallocation auctions requires alignment between preference substitutability and the direction of efficient trades. Package substitutability is useful only when the efficient reallocation looks like a one-sided allocation of packages.

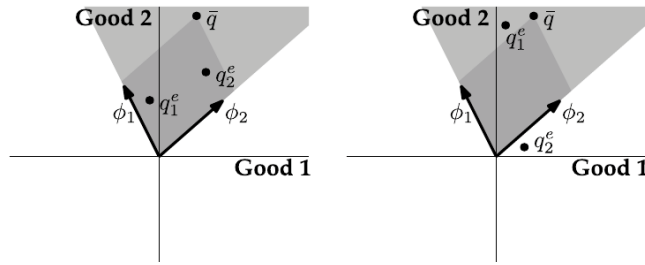


Figure 1: Theorem 1's allocation condition

5.2 Figure 2: Allocations and core-selection conditions in Examples 1–3

Read:

- The first panel corresponds to Example 1. With a single good, no package basis can make all three bidders receive positive package allocations.
- The second panel corresponds to Example 2. Efficient allocations lie in the cone generated by the packages $\phi_1 = [1, 1]'$ and $\phi_2 = [-1, 1]'$.
- The third panel corresponds to Example 3. A subset of bidders generates an efficient allocation outside the positive orthant where Theorem 1's allocation condition would hold for substitutable goods.
- The panels show the three possible lessons: core selection can fail with one good, succeed with complementarities, and fail with substitutable goods.

Interpretation. The figure summarizes why the paper's result is not a small modification of the one-sided gross-substitutes theorem. Examples 1 and 3 show that reallocation trade can force subsidized bidder-to-bidder transfers. Example 2 shows that complementarity among named goods is not fatal if the correct package basis exists.

Economic message. The figure is the paper’s conceptual map. The decisive issue is whether the efficient allocation can be expressed as positive allocations of common substitute packages, not whether the original goods are substitutes.

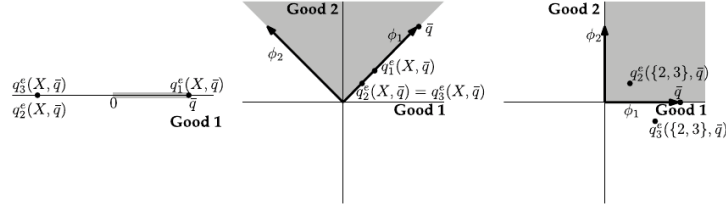


Figure 2: Allocations and core-selection conditions in examples

6 Interpretive synthesis

6.1 What the paper adds to core-selecting auction theory

The classical result is that gross substitutability supports core selection in one-sided auctions. Rostek and Yoder show that reallocative auctions require a different criterion. Once bidders can sell as well as buy, the surplus function includes gains from trade among bidders, not just gains from trade with the auctioneer. The Vickrey auction, because it internalizes those gains efficiently, may have to subsidize them.

6.2 Why heterogeneity is the central object

Heterogeneity creates efficient bidder-to-bidder trades. There are two independent dimensions:

- heterogeneity in pre-auction marginal utilities, which means the initial allocation itself contains gains from trade;
- heterogeneity in substitution patterns, which means new supply from the auctioneer can create further gains from trading other goods among bidders.

When heterogeneity is small, most surplus comes from the auctioneer’s quantity. When heterogeneity is large, much surplus comes from reallocation among bidders, which Vickrey subsidizes.

6.3 Why packages are more than a bidding convenience

Packages are not just a practical way to simplify bidding. In the theory, packages reveal the correct basis in which preferences behave like substitutes. If designers can identify eigenvector-like packages, they can both diagnose core selection and simplify implementation.

6.4 Why Vickrey can be good or bad

Vickrey is efficient and incentive compatible, but in reallocative environments efficiency can require subsidized reallocation. Thus Vickrey performs well when heterogeneity is limited and poorly when heterogeneity is large. The uniform-price auction sacrifices efficiency but avoids some subsidy exposure.

6.5 Design implication

In applications such as spectrum, water rights, or electricity contracts, designers should ask:

- Are participants mainly reallocating pre-existing holdings, or is the auctioneer introducing new quantity that creates surplus?
- Do bidders have similar substitution patterns?
- Can package directions be chosen to align with common substitution patterns?
- Is revenue competitiveness more important than full allocative efficiency?

7 Short summary

- The paper studies core selection in reallocative auctions, where bidders can both buy and sell goods.
- In one-sided auctions, gross substitutability is enough for core selection; in reallocative auctions, it is neither necessary nor sufficient.
- The relevant core notion is often the bidder-core, which focuses on revenue competitiveness while allowing that revenue may be negative.
- The Vickrey auction is the benchmark because incentive-compatible core-selecting mechanisms must be Vickrey-equivalent on convex domains.
- Example 1 shows core selection can fail even with one good and positive revenue.
- Example 2 shows core selection can hold even when some bidders view goods as complements.
- Example 3 shows core selection can fail even when goods are substitutes for all bidders.
- Theorem 1 gives sufficient conditions: there must be substitute packages, and efficient allocations must give every bidder nonnegative quantities of those packages.
- Theorem 2 gives primitive conditions: equal pre-auction marginal utilities and commuting substitution-pattern matrices imply a common eigenbasis of packages and core selection.
- Propositions 3–5 show that these homogeneity conditions are necessary in a maximal-domain sense.
- Theorem 3 compares Vickrey and uniform-price revenue: Vickrey wins when expected heterogeneity is low relative to the auctioneer’s quantity; UPA can win when heterogeneity is high.
- Proposition 7 shows that eigenvector packages can decentralize a complex joint auction into independent package auctions.
- Proposition 8 shows that splitting a Vickrey auction into forward and reverse auctions changes incentives and destroys dominant-strategy truthful bidding.
- The central lesson is that in reallocative auctions, revenue competitiveness depends on bidder heterogeneity and the direction of efficient reallocation.

8 Conclusion

8.1 Core conclusion

Rostek and Yoder show that the logic of core selection changes fundamentally when auctions reallocate goods among participants rather than merely sell goods from an auctioneer to buyers. Substitutability among original goods is no longer the right criterion. What matters is whether bidders are sufficiently homogeneous in pre-auction marginal utilities and substitution patterns, so that efficient reallocation can be described as positive allocations of common substitute packages.

8.2 Economic intuition

A Vickrey auction pays each bidder his marginal contribution. In a one-sided sale, this often yields competitive revenue under gross substitutes. In a reallocative auction, however, efficiency may require moving goods among bidders. If these trades mostly reflect pre-existing gains from trade between heterogeneous bidders, the auctioneer may subsidize them. This is why core selection can fail even with substitutable goods and positive revenue.

8.3 Policy and design implications

The paper’s message for market designers is practical. If the design problem is mainly to add new supply and bidders have similar substitution patterns, core-selecting Vickrey-style designs can perform well. If the problem is mainly to reallocate goods among highly heterogeneous participants, uniform-price rules or other designs may be more attractive. When package bidding is needed, designers should focus on package directions that reflect common substitution patterns, such as buying packages and switching packages.

8.4 Final takeaway

The paper replaces a goods-based view of core selection with a heterogeneity-and-packages view. In reallocative markets, the designer must understand not only what goods are traded, but which bidder-to-bidder trades efficiency requires and whether those trades can be represented as one-sided allocations of common substitute packages.